Chapter II.7, Week 9

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7.1. Let $(X, \mathcal{O}_X)$ be a locally ringed space, and let $f : \mathcal{L} \to \mathcal{M}$ be a surjective map of invertible sheaves on X. Show that f is an isomorphism. [*Hint:* Reduce to a question of modules over a local ring by looking at the stalks.]

Proof. It suffices to show $f_x : \mathcal{L}_x \twoheadrightarrow \mathcal{M}_x$ is an isomorphism of $\mathcal{O}_{X,x}$-modules. $\mathcal{L}$, $\mathcal{M}$ invertible implies $\mathcal{L}_x \cong \mathcal{M}_x \cong \mathcal{O}_{X,x}$, so it suffices to show that a surjective map $f : \mathcal{O}_{X,x} \twoheadrightarrow \mathcal{O}_{X,x}$ is an isomorphism. We have $a \in \mathcal{O}_{X,x}$ such that $f(a) = 1$, so $f(1) = a^{-1}$ is a unit, whence $0 = f(b) = bf(1)$ implies $b = 0$, showing injectivity. $\square$

7.2. Let X be a scheme over a field k. Let $\mathcal{L}$ be an invertible sheaf on X, and let $\{s_0, \dots, s_n\}$ and $\{t_0, \dots, t_m\}$ be two sets of sections of $\mathcal{L}$, which generate the same subspace $V \subset \Gamma(X, \mathcal{L})$, and which generate the sheaf $\mathcal{L}$ at every point. Suppose $n \leqslant m$. Show that the corresponding morphisms $\phi : X \to \mathbb{P}^n_k$ and $\psi : X \to \mathbb{P}^m_k$ differ by a suitable linear projection $\mathbb{P}^m - L \to \mathbb{P}^n$, and an automorphism of $\mathbb{P}^n$, where L is a linear subspace of dimension $m - n - 1$.

7.5. Establish the following properties of ample and very ample invertible sheaves on a noetherian scheme X. $\mathcal{L}$, $\mathcal{M}$ will denote invertible sheaves, and for (d), (e) we assume furthermore that X is of finite type over a noetherian ring A.

(a) If $\mathcal{L}$ is ample and $\mathcal{M}$ is generated by global sections, then $\mathcal{L} \otimes \mathcal{M}$ is ample.
(b) If $\mathcal{L}$ is ample and $\mathcal{M}$ is arbitrary, then $\mathcal{M} \otimes \mathcal{L}^n$ is ample for $n \gg 0$.
(c) If $\mathcal{L}$, $\mathcal{M}$ are both ample, so is $\mathcal{L} \otimes \mathcal{M}$.
(d) If $\mathcal{L}$ is very ample and $\mathcal{M}$ is generated by global sections, then $\mathcal{L} \otimes \mathcal{M}$ is very ample.
(e) If $\mathcal{L}$ is ample, then there is an $n_0 > 0$ such that $\mathcal{L}^n$ is very ample for all $n \geqslant n_0$.

Proof. (a) Let $\mathcal{F}$ be a coherent sheaf on X, take global generators $s_0, \dots, s_r \in \Gamma(X, \mathcal{M})$, and let $n_0 > 0$ such that $\mathcal{F} \otimes \mathcal{L}^n$ is globally generated for all $n \geqslant n_0$. Then, for any $n \geqslant n_0$, $\mathcal{M}^n$ is globally generated by the $s_{i_1} \otimes \cdots \otimes s_{i_n} \in \Gamma(X, \mathcal{M})$. Choosing some generators $t_0, \dots, t_k \in \Gamma(X, \mathcal{F} \otimes \mathcal{L}^n)$, we have generators

$$t_{i_0} \otimes s_{i_1} \otimes \cdots \otimes s_{i_n} \in \Gamma(X, \mathcal{F} \otimes (\mathcal{L} \otimes \mathcal{M})^n).$$

These sections do indeed generate, because $(\mathcal{F} \otimes (\mathcal{L} \otimes \mathcal{M})^n)_x \cong \mathcal{F}_x \otimes (\mathcal{L}_x \otimes \mathcal{M}_x)^n$.

(b) Let $n_0 > 0$ such that $\mathcal{M} \otimes \mathcal{L}^n$ is globally generated whenever $n \geqslant n_0$. We will show that $\mathcal{M} \otimes \mathcal{L}^{n_0+1}$ is ample. Fix $\mathcal{F} \in \mathrm{Coh}_X$, and take $n_1 > 0$ such that $\mathcal{F} \otimes \mathcal{L}^n$ is globally generated for all $n \geqslant n_1$. Then, for any $n \geqslant n_1$,

$$\mathcal{F} \otimes (\mathcal{M} \otimes \mathcal{L}^{n_0+1})^n \cong (\mathcal{F} \otimes \mathcal{L}^n) \otimes (\mathcal{M} \otimes \mathcal{L}^{n_0})^n$$

is a tensor product of globally generated spaces, hence globally generated itself.

(c) Fix $\mathcal{F} \in \mathrm{Coh}_X$, take $n_0 > 0$ such that $\mathcal{F} \otimes \mathcal{L}^n$ is globally generated whenever $n \geqslant n_0$, and take $n_1 > 0$ such that $\mathcal{M} \otimes \mathcal{L}^n$ is globally generated whenever $n \geqslant n_1$. Then, whenever $n \geqslant \max\{n_0 + 1, n_1\}$, we have that

$$\mathcal{F} \otimes (\mathcal{M} \otimes \mathcal{L})^n = (\mathcal{F} \otimes \mathcal{M}^{n-1}) \otimes (\mathcal{M} \otimes \mathcal{L}^n)$$

is globally generated.

(d) Take generators $s_0, \dots, s_m \in \Gamma(X, \mathcal{L})$ and $t_0, \dots, s_n \in \Gamma(X, \mathcal{M})$, along with corresponding maps $\phi : X \to \mathbb{P}^m$ and $\psi : X \to \mathbb{P}^n$ such that

$$\mathcal{L} \cong \phi^*(\mathcal{O}_{\mathbb{P}^m}(1)), \quad \mathcal{M} \cong \psi^*(\mathcal{O}_{\mathbb{P}^n}(1)).$$

Consider the Segre embedding $\sigma : \mathbb{P}^m \times \mathbb{P}^n \to \mathbb{P}^N$, and the composition

$$X \xrightarrow{\Delta} X \times X \xrightarrow{\phi \times \psi} \mathbb{P}^m \times \mathbb{P}^n \xrightarrow{\sigma} \mathbb{P}^N.$$

Denoting the above map $f : X \to \mathbb{P}^N$, we have

$$\begin{aligned} f^*(\mathcal{O}_{\mathbb{P}^N}) &\cong ((\phi \times \psi) \circ \Delta)^*(p_1^*(\mathcal{O}_{\mathbb{P}^m}(1)) \otimes p_2^*(\mathcal{O}_{\mathbb{P}^n}(1))) \\ &\cong \phi^*(\mathcal{O}_{\mathbb{P}^n}(1)) \otimes \psi^*(\mathcal{O}_{\mathbb{P}^m}(1)) \\ &\cong \mathcal{L} \otimes \mathcal{M} \end{aligned}$$

by (Ex. 5.11). It suffices, then, to show that $g := (\phi \times \psi) \circ \Delta$ is a closed immersion.

By (7.2), we have each $X_i := X_{s_i}$ is affine, with each

$$\begin{aligned} A[z_0, \dots, z_m] &\longrightarrow \Gamma(X_i, \mathcal{O}_{X_i}) \\ z_j &\longmapsto s_j/s_i \end{aligned}$$

surjective, where $z_j := x_j/x_i$. Consider $Y_j := X_{t_j}$.Then $X_i \cap Y_j = D_{X_i}(t_j)$ is affine, so surjectivity of the map

$$\begin{aligned} A[w_0, \dots, w_n] &\longrightarrow \Gamma(Y_j, \mathcal{O}_{Y_j}) \\ w_k &\longmapsto t_k/t_j \end{aligned}$$

gives us surjectivity of the restricted map $A[w_0, \dots, w_n] \to \Gamma(X_i \cap Y_j, \mathcal{O}_{X_i \cap Y_j})$. Hence, the map

$$\begin{aligned} A[z_0 w_0, \dots, z_m w_n] &\longrightarrow \Gamma(X_i \cap Y_j, \mathcal{O}_{X_i \cap Y_j}) \\ z_k w_l &\longmapsto \frac{x_k y_l}{x_i y_j} \end{aligned}$$

is surjective.

We have $X_i \cap Y_j = g^{-1}(U_{ij}) = g^{-1}(D_+(x_i y_j))$, so that g is locally a closed immersion on the base. We conclude that, in fact $g : X \to \mathbb{P}^m \times \mathbb{P}^n$ is a a closed immersion, whence $\sigma \circ g : X \to \mathbb{P}^N$ is a closed immersion with $\mathcal{L} \otimes \mathcal{M} \cong (\sigma \circ g)^*(\mathbb{P}^N)$. That is, $\mathcal{L} \otimes \mathcal{M}$ is very ample with respect to A.

(e) Take some fixed $m > 0$ such that $\mathcal{L}^m$ is very ample by (7.6), and take $n_0 > 0$ such that $\mathcal{O}_X \otimes \mathcal{L}^n \cong \mathcal{L}^n$ is globally generated for all $n \geqslant n_0$. Then, for all $n \geqslant m + n_0$, we have that $\mathcal{L}^n \cong \mathcal{L}^m \otimes \mathcal{L}^{n-m}$ is very ample by (d). □

7.7. *Some Rational Surfaces.* Let $X = \mathbb{P}^2_k$, and let $|D|$ be the complete linear system of all divisors of degree 2 on X (conics). D corresponds to the invertible sheaf $\mathcal{O}(2)$, whose space of global sections has a basis $x^2, y^2, z^2, xy, xz, yz$, where x, y, z are the homogneous coorrdinates of X.

(a) The complete linear system $|D|$ gives an embedding of $\mathbb{P}^2$ in $\mathbb{P}^5$, whose image is the veronese surfaces (I, Ex. 2.13).

(b) Show that the subsystem defined by $x^2, y^2, z^2, y(x-z), (x-y)z$ gives a closed immersion of X into $\mathbb{P}^4$. The image is called the *Veronese surface* in $\mathbb{P}^4$. Cf. (IV, Ex. 3.11).

(c) Let $\mathfrak{d} \subset |D|$ be the linear system of all conic passing through a fixed point p. Then $\mathfrak{d}$ gives an imersion of $U = X - p$ into $\mathbb{P}^4$. Furthermore, if we blow up p to get a surface $\widetilde{X}$, then this map extends to give a closed immersion of $\widetilde{X}$ in $\mathbb{P}^4$. Show that $\widetilde{X}$ is a surface of degree 3 in $\mathbb{P}^4$, and that the lines in X through p are transformed into straight lines in $\widetilde{X}$ which do not meet. $\widetilde{X}$ is the union of all these lines, so we say $\widetilde{X}$ is a *ruled surface* (V, 2.19.1).